

Unit 12, Lesson 6: Discovering Inverse Operations Equation

Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.

Learning Target

- ☐ I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations.

Dale is helping to fix the roof at his uncle's house. They are using a 40-foot ladder. Dale is climbing the ladder to reach the roof. He is currently 7 feet up the ladder.

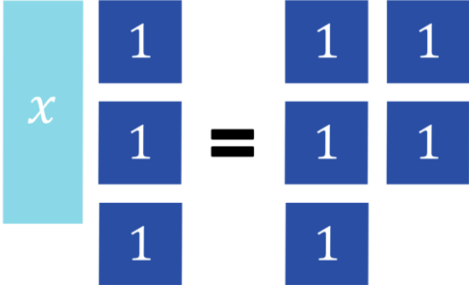
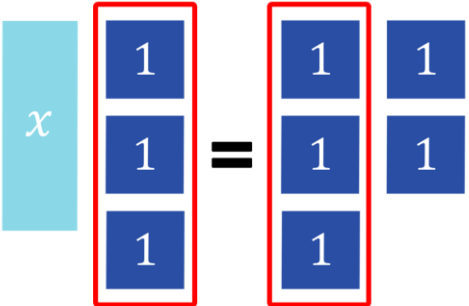
1. Sketch a picture to represent this situation.
2. How much farther does Dale have to go to reach the roof? Include units.

12.6.2 Exploration: Solving Equations Using Algebra Tiles

Algebra tiles can be used to represent and solve algebraic equations.

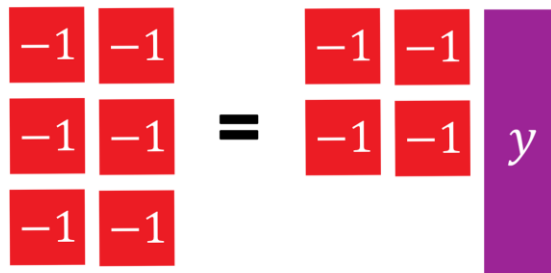
In the model shown,  represents 1 and  represents x .

1. The steps to determine the value of  using algebra tiles are shown.
- a. Work with your partner to complete the last step according to the description.

Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

- b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.
- c. How many blue tiles are equivalent to one teal tile?
- d. What is the value of x in the equation $x + 3 = 5$?

2. In the model shown,  represents -1 and  represents y



$$\begin{array}{|c|c|} \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline \end{array} = \begin{array}{|c|c|} \hline -1 & -1 \\ \hline -1 & -1 \\ \hline \end{array} y$$

- a. Work with your partner to determine how many red tiles are equivalent to one purple tile.
- b. What is the value of y in the equation $-6 = -4 + y$?

12.6.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

- 1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

When finding the solution to an equation, inverse operations are used to balance an equation.

An **inverse operation** is an operation that reverses the effect of another operation.

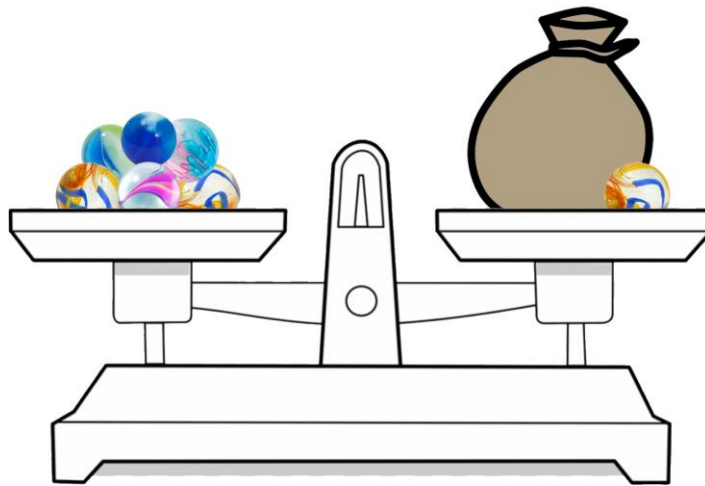
- 2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

Operation	Inverse Operation
Subtraction (−)	
Addition (+)	

12.6.4 Guided Practice: Using Inverse Operations to Solve Equations

Balance scales are used to measure weight, or mass. When both pans are at the same height, that means the objects in the two pans weigh the same amount. In mathematics, balance scales are often used to show balanced equations where the expressions on either side of the balance are equivalent.

1. Heidi and Pete were given the balance scale shown below, which has exactly 6 marbles in the left pan. Heidi and Pete were asked how many marbles are in one bag.

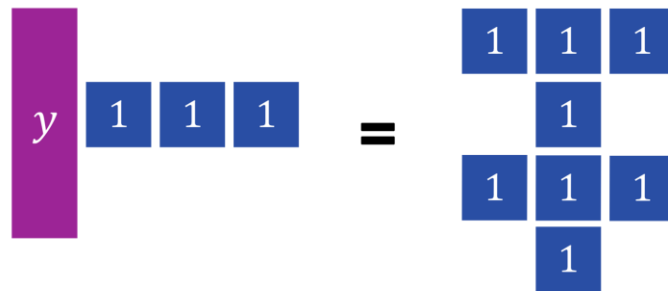


Heidi stated, "There is not enough information to solve this problem."	Pete stated, "Each bag holds 5 marbles."
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- a. Explain who is correct.
- b. Write an algebraic equation to represent the balance scale problem Heidi and Pete were given.
- c. How many marbles are in one bag?

d. What inverse operation can be used to solve the equation?

2. The algebra tiles shown are used to model an equation.



a. How many blue tiles are equivalent to one purple tile?

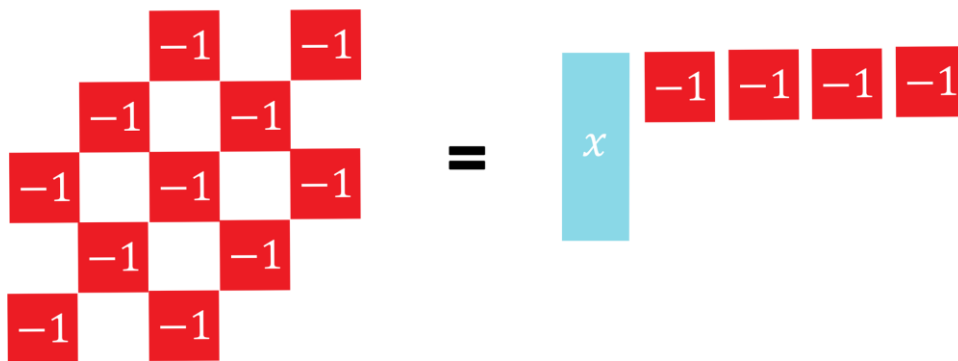
b. Write a one-variable equation to represent the tiles on the mat.

c. What inverse operation can be used to solve the equation?

d. What value of y makes the equation true?

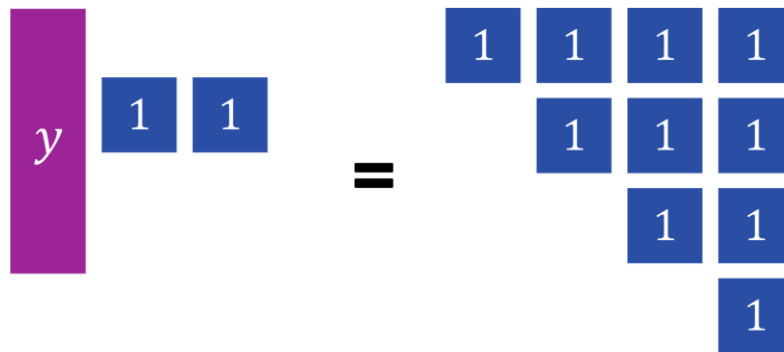
12.6.5 Your Turn!

1. The algebra tiles shown are used to model an equation.



- How many red tiles are equivalent to one teal tile?
- Write a one-variable equation to represent the tiles on the mat.
- What inverse operation can be used to solve the equation?
- What value of x makes the equation true?

2. The algebra tiles below model an equation.



- How many blue tiles are equivalent to one purple tile?
- Write a one-variable equation to represent the tiles on the mat.
- What inverse operation can be used to solve the equation?
- What value of y makes the equation true?

12.6.6 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class summarizing today's lesson.